



STATISTICAL LEARNING FOR AUTOMATION SYSTEMS

1. The feasibility of learning

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Prof. Simone Formentin

THE LEARNING PROBLEM

INPUT: $x \in X$

$x = [x_1, x_2, x_3, \dots, x_n]^T$
a set of email attributes

OUTPUT: $y \in Y$

$y = [\text{spam}/\text{ham}]$

TARGET FUNCTION:

$$f: X \rightarrow Y$$

"ideal" law that maps
email attributes onto a certain y

DATA: $D_N = \{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(N)}, y^{(N)})\}$

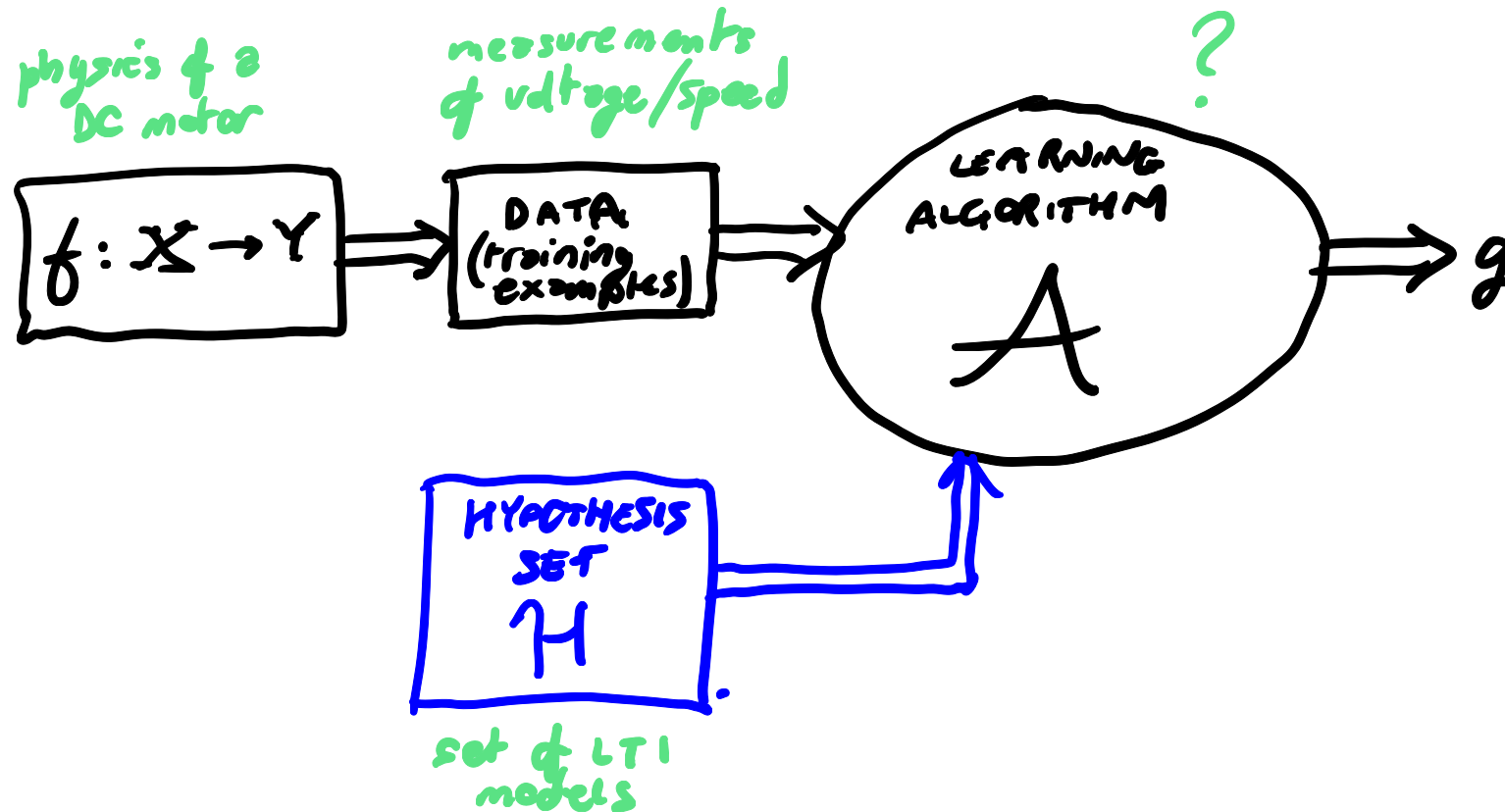
set of emails in the dataset

HYPOTHESIS: $g: X \rightarrow Y, g \in \mathcal{H}$

candidate model

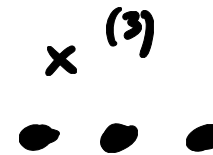
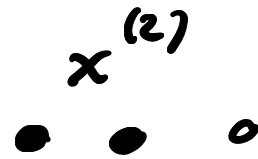
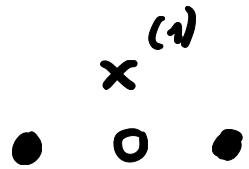
HYPOTHESIS SET: $\mathcal{H}$

THE LEARNING PROCEDURE

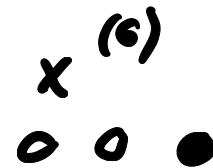
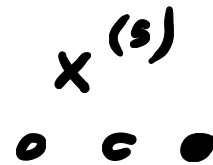
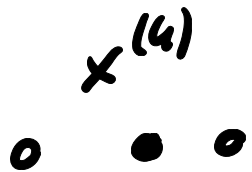


LEARNING FROM DATA IS MEANINGFUL
IF

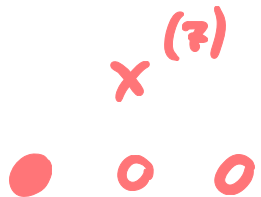
- ① A PATTERN EXISTS
- ② WE CANNOT PIN IT DOWN MATHEMATICALLY
- ③ WE HAVE DATA



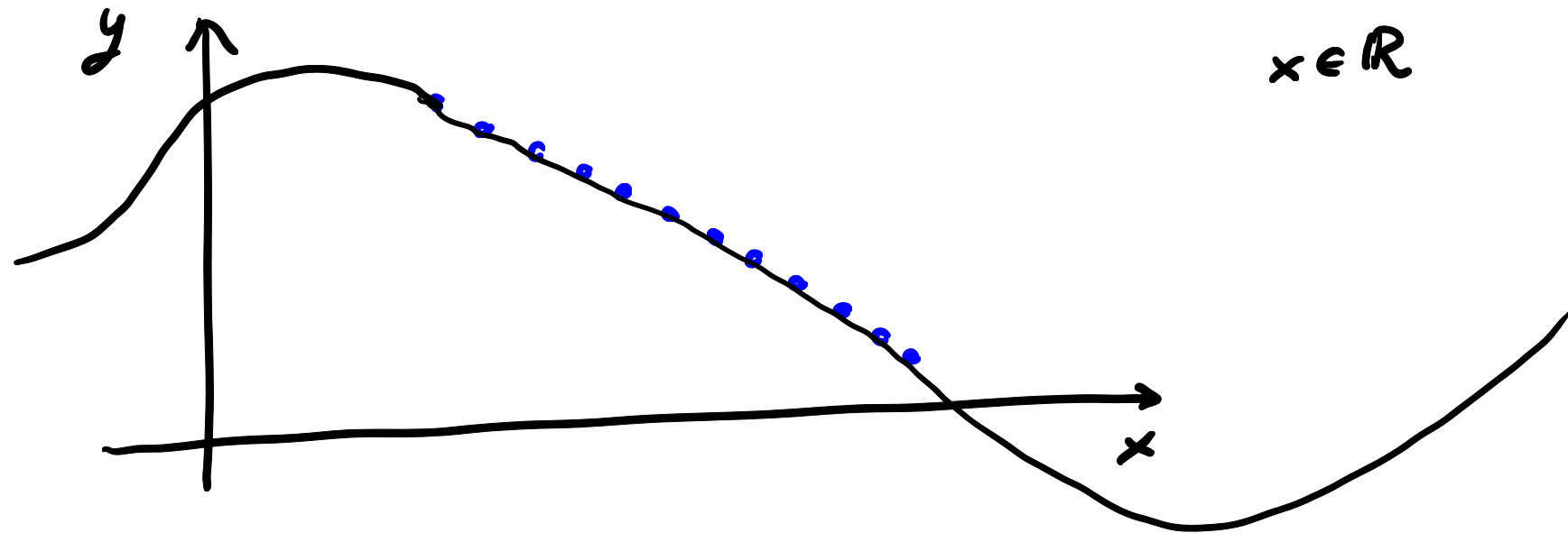
$$x^{(1)}, x^{(2)}, x^{(3)} \\ \Downarrow \\ y = +1$$



$$x^{(4)}, x^{(5)}, x^{(6)} \\ \Downarrow \\ y = -1$$



$$y = ?$$



NO, learning from data is NOT feasible

(Hume's problem of induction, 1748)

... but only from a deterministic point of view!

It is possible from a statistical point of view!

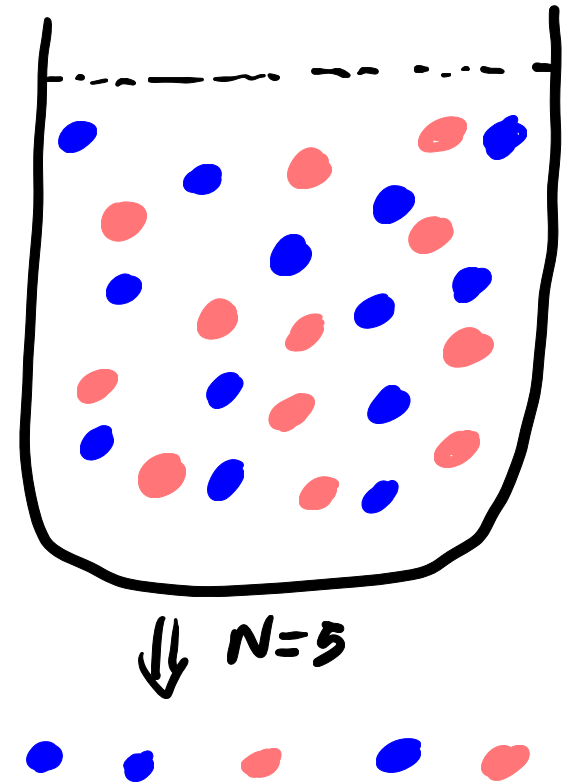
BIN EXPERIMENT

- we pick N marbles independently

p : $P[\text{picking a blue marble}]$

$\hat{p}$: fraction of blue marbles in the sample

- Does $\hat{p}$ say anything about p ?



No, it is possible that the sample is mostly red,
while the bin is mostly blue

(PESSIMISTIC perspective)

Yes, if the sample is large, it is probable that $\hat{p}$ is close to p
(OPTIMISTIC perspective)

The mathematical support is the law of large numbers!

Hoeffding's inequality

$$P[\hat{p} \text{ being far from } p] \leq b(N)$$

$$P[|\hat{p} - p| > \epsilon] \leq 2e^{-2\epsilon^2 N}$$

- Trade-off between the accuracy ϵ and the size of the experiment N
- If $\epsilon \rightarrow 0$, $2e^{-2\epsilon^2 N} \rightarrow 1$ ($P[\cdot] \leq 2$, useless)
- $\hat{p} = p$ P.A.C. statement $\rightarrow$ P: probably
 $\rightarrow$ A: approximately
 $\rightarrow$ C: correct

CONNECTION TO THE LEARNING PROBLEM

Bin

The unknown is p

Inputs: each marble

Learning problem (classification)

The unknown is the target function f

Input: each x

Given g

blue marble $\rightarrow g(x) \neq f(x)$

red marble $\rightarrow g(x) = f(x)$

p could be seen as the fraction of misclassified points overall
 $\hat{p}$ could be seen as the fraction of misclassified points in-sample

IN-SAMPLE ERROR

$$E_{in}(g) = \frac{1}{N} \sum_{i=1}^N \llbracket g(x_i) \neq f(x_i) \rrbracket$$

$$\begin{aligned} \llbracket \cdot \rrbracket &= 1 && \text{if the statement is true} \\ &= 0 && \text{otherwise} \end{aligned}$$

OUT-OF-SAMPLE ERROR

$$E_{out}(g) = P[g(x) \neq f(x)]$$

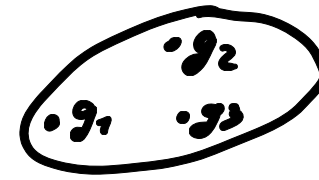
$$P[|\hat{p} - p| > \epsilon] \leq 2e^{-2\epsilon^2 N} \quad \text{bin experiment}$$

$$P[\underbrace{|E_{in}(g) - E_{out}(g)|}_{\text{bad event}} > \epsilon] \leq \boxed{2e^{-2\epsilon^2 N}} \quad \begin{array}{l} \text{verification} \\ \text{of a hypothesis } g(\cdot) \end{array}$$

↓
does not depend on E_{out} !

Not yet "learning"....

g is NOT given



H

example;
 $M=3$ candidate models

$$P[|E_{in}(g) - E_{out}(g)| > \epsilon] \leq 2M e^{-2\epsilon^2 N}$$

M : number of candidate models

OBJECTIVES OF LEARNING

FITTING

① minimize $E_{in}(g)$ with respect to $g \in \mathcal{M}$

GENERALIZATION

② minimize $|E_{in}(g) - E_{out}(g)|$

RECALL THAT

minimization of $E_{out}(g)$
is NOT possible!